

Muhammad Saad Imran

Full-Stack & AI Developer • React, FastAPI & Node • RAG, Agents and LLM Applications

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PROFILE

I take products from concept to deployed — data model, REST API, authentication, file handling, the interface people actually use, and the AI layer on top. My AI work is application engineering rather than prompting: retrieval over a vector store with typed outputs and a real model-fallback chain, and an agent whose permissions make an irreversible action impossible rather than discouraged. I have a gaming platform live in production on Node and MongoDB, a JavaFX desktop application documented well enough for a stranger to clone and run with one command, and a retrieval service in FastAPI behind a React front end. I test what I build, and I write down the limitations. Final-year Software Engineering student and a recent machine learning engineering intern at FlyRank AI.

EXPERIENCE

Machine Learning Engineering Intern — FlyRank AI

Jul – Sep 2026

Remote • Sarajevo, Bosnia and Herzegovina • Python • DuckDB over cloud Parquet • scikit-learn • GitHub Actions CI

- Built a page-ranking model on an **81-million-row** search warehouse, queried in place rather than downloaded. Precision@50 rose from **0.492 to 0.784** against a 0.216 base rate, under a client-grouped, time-aware split.
- Reported the caveats beside the result: roughly half the gain rides on click history, and a deliberate leakage test moved R^2 from an honest 0.037 to a flattering 0.874 before the offending column was removed.
- Audited my own headline claim about the validation split and found it inflated; published the corrected, smaller number instead.
- Shipped the capstone research paper as a self-contained page, audited against the Pineau reproducibility checklist, arguing against its own headline where the evidence did.
- 32 practical assignments • machine learning track complete at 10 / 10 • capstone reviewed and accepted by the lead track mentor.

Certificate of Completion, credential **FR-D11-F1DDA-CDAA7** — publicly verifiable at internship.flyrank.ai

SELECTED PROJECTS

Bongupid Games — browser gaming & publishing platform

2026 • **Live**

Node.js • Express • MongoDB / Mongoose • REST • JWT • Google OAuth • Cloudflare R2 • Vanilla ES6 SPA

- Designed and shipped the whole stack: an Express REST API with JWT auth, Google sign-in, OTP password recovery and role-separated admin endpoints, over a Mongoose model covering games, users, ratings, comments, history and analytics.
- Built the publishing pipeline developers actually use — multipart upload that validates and extracts ZIP game builds into playable routes, cover art and screenshots on Cloudflare R2, visibility and release-status controls, and a per-game analytics view.
- Wrote the client as a **2,451-line single-page app in vanilla JavaScript** — discovery, search, in-browser play, ratings and reviews, no framework — and hardened it for the open web with rate limiting, bot protection and SEO controls.
- Live at submission with 15 published builds and 11 active users.

Live: bongupid.games • two-person build

DecisAI — AI bid-response & RFP decision system

CUST Hackathon 2026

React 19 • Vite • FastAPI • Python • ChromaDB • Sentence Transformers • scikit-learn • Claude API • Ollama

- End-to-end AI product: a React front end over a FastAPI backend of 11 documented endpoints — upload, structured extraction, evidence matching, compliance classification, draft generation, win scoring, GO / NO-GO and export to DOCX or PDF.
- Built the retrieval layer that matches each requirement in a tender against the company's real delivery record, returning scored evidence rather than prose, and flagging **"no capability match"** instead of drafting a plausible answer.
- A real fallback chain: Claude API when configured, local Ollama by default so client documents never leave the network, and a deterministic template path when neither is available — so the product degrades instead of failing.
- Ran a **500-page tender end to end** — 91 requirements extracted at 94% confidence, each checked against evidence, unsupported ones rendered as an explicit evidence gap.

Repo: github.com/Saad-Imran-Toori/DecisAI

AssetForge Studio — JavaFX desktop workspace for game studios

2026 · Public

Java 21 · JavaFX 21 · Maven · MySQL 8 + HikariCP · JasperReports · JUnit 5 · H2

- Seven modules in one desktop application — dashboard, projects, phases, a drag-and-drop Kanban board, an asset library, a bug tracker and a PDF report generator — across **100 source files**.
- Layered MVC with separate Service and DAO tiers, controllers depending on interfaces so new report types and board statuses are additive. Singleton, DAO, Observer, Factory, Strategy and Builder, used deliberately.
- Kept phases, tasks, bugs, assets and the dashboard consistent through event-driven synchronisation over an EventBus; asset checksums drive duplicate and missing-file detection.
- Ships a zero-setup demo mode on in-memory H2 — one Maven command, no database — plus four themes including a high-contrast palette meeting WCAG 2.2 AA.

Repo: github.com/Saad-Imran-Toori/AssetForge-Studio · two-person build

RepoWiz — AI agent with structural guardrails

2026

Agent design · tool permissioning · evaluation-first development

- An agent that gathers its own evidence and marks every claim verified with a source, or unverified with a question back.
- Six evaluation cases were written before the build. On live runs it refused a figure it could not source, and found two real errors in my own records unprompted.
- Guardrails are structural, not instructed — no write, send or publish tool exists, so an irreversible action is impossible rather than discouraged.

Also built: a multi-layer automated test framework for an e-commerce admin panel — Selenium WebDriver & Grid in Docker, JUnit 5, PHPUnit and Postman · a complete two-role bus-booking application designed in Figma across 25+ linked screens for an external client · an unbeatable Tic-Tac-Toe AI using Minimax with depth-based evaluation, rendered entirely from Pygame primitives.

TECHNICAL SKILLS

AI & LLM	Anthropic Claude API · RAG pipelines · vector databases (ChromaDB) · embeddings (Sentence Transformers) · structured and typed AI outputs · model fallback architecture · local inference with Ollama · agent design and tool permissioning · prompt engineering and cross-model evaluation
Backend	Python · FastAPI · uvicorn · Node.js · Express · REST API design and contracts · authentication and permissions (JWT, Google OAuth, OTP) · multipart upload and file handling · rate limiting and bot protection
Frontend	React 19 · Vite · JavaScript (ES6) · HTML5 · CSS3 · single-page architecture · responsive design · dashboards and data visualisation · Figma and Google Stitch for UI design
Data	Relational modelling (MySQL, SQL) · MongoDB / Mongoose · DuckDB over Parquet · scikit-learn · pandas · NumPy · grouped and time-aware validation · leakage detection
Desktop	Java 21 · JavaFX · FXML · Maven · HikariCP · JasperReports · MVC / Service / DAO layering · design patterns
Delivery	Git & GitHub · GitHub Actions CI · Docker · cloud deployment (Cloudflare R2, GitHub Pages) · JUnit 5 · Mockito · Selenium WebDriver & Grid · PHPUnit · Postman · technical documentation · Jira (Agile)

EDUCATION

BS Software Engineering — Capital University of Science & Technology, Islamabad

Expected 2027

CGPA 3.66 / 4.00 · final-year student

Relevant coursework: Artificial Intelligence · Web Engineering · Software Design & Architecture · Software Construction & Development · Database Systems · Automated Software Testing

CERTIFICATIONS

FlyRank AI	Certificate of Completion, Machine Learning Internship · credential FR-D11-F1DDA-CDAA7 (Sep 2026) · recommendation letter from the Founder & CEO
Anthropic Academy	Six courses, including AI Fluency: Framework & Foundations and Introduction to Agent Skills (Jul – Aug 2026)
Google (Coursera)	AI Essentials and Prompting Essentials specializations, plus six further AI courses (Jul – Aug 2026)
Atlassian (Coursera)	Agile with Atlassian Jira (Jul 2026)